CSCI 6840
Project 1

Project Assignment: Customer Segmentation Using Clustering Techniques

Objective:
The goal of this project is to apply clustering techniques to perform customer segmentation on a dataset. You will explore different clustering algorithms, evaluate their performance, and interpret the clusters to draw meaningful business insights.

Dataset:
You will use the "Customer Segmentation" dataset (the Mall Customers dataset from Kaggle). This dataset contains demographic information, purchasing behavior, and other relevant data about customers.

Project Steps:

1. **Data Exploration and Preprocessing:**
   - Load the dataset and perform exploratory data analysis (EDA).
   - Handle missing values, if any.
   - Normalize or standardize the data as needed.
   - Select relevant features for clustering (e.g., age, annual income, spending score).
2. **Clustering Techniques:**
   - Apply at least two different clustering algorithms (e.g., K-Means, fuzzy K-Means).
   - Determine the optimal number of clusters using methods like the elbow method, silhouette score, or dendrogram analysis.
   - Compare and contrast the results of different clustering algorithms.
3. **Cluster Analysis:**
   - Visualize the clusters using appropriate techniques (e.g., scatter plots).
   - Interpret the characteristics of each cluster. Describe the common features of customers within the same cluster.
   - Provide business insights based on the clusters (e.g., targeted marketing strategies for different customer segments).
4. **Evaluation:**
   - Evaluate the quality of the clusters using metrics like silhouette score.
   - Discuss the strengths and limitations of each clustering algorithm used.
5. **Report and Presentation:**
   - Prepare a detailed report that includes your data exploration, methodology, results, and conclusions.
   - Create a presentation that summarizes your findings and business recommendations.

Deliverables:

- Python code implementing clustering analysis. Show screenshot to demonstrate that your program is implemented and executed on the Bridges2-GPU server at PSC.
- A written report (4-6 pages) explaining your approach, results, and insights. In the report, CLEARLY list what your contributions are.
- A group presentation (10+ slides) summarizing your project findings.

Grading Criteria:

- Data preprocessing: 20%
- Implementation of clustering algorithms: 30%
- Quality and interpretation of clusters: 30%
- Report and presentation: 20%

Additional Notes:

- Extra credit will be given for the application of advanced clustering techniques (e.g., Gaussian Mixture Models, Spectral Clustering) or innovative visualizations.
- Collaborate with your group peers to discuss potential challenges and solutions, but ensure that your work is original.